Michael "Fitz" Fitzgerald

Senior Software Engineer | Applied AI • React/TypeScript • Compliance & Governance

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Clearance: U.S. Citizen. Held multiple Public Trust Clearances (CPSC, HHS/HRSA); highly clearable and active-ready for federal contract sponsorship.

SUMMARY

Lately I've been focused on spec-driven development: building *reliable* AI agents around highly detailed instructions with objectively measurable goals, human-in-the-loop checkpoints, and copious guardrails to keep them from going rogue.It's the same governance discipline I've brought to 20+ years of shipping applications for federal agencies and enterprise systems. Apps that are maintainable and accessible such as HRSA's multi-site Angular platform and, more recently, CPSC's internal systems rebuilt with React. Specialized in 508 and regulatory compliance implementation, legacy system refactoring, and translating complex stakeholder requirements into clean, scalable code. My unique combination of frontend engineering depth and UX sensibility means I build systems that are both technically sound and genuinely usable.

CORE COMPETENCIES

- **AI Engineering:** AI Orchestration, Multi-Agent Patterns, Human-in-the-Loop Design, Guardrails, Prompt Engineering, RAG
- **AI Quality & Governance:** Evaluation Frameworks, Benchmarking, Spec-Driven Development, AI Governance
- **Compliance & Enterprise:** Regulatory Compliance, Section 508 / WCAG 2.1 AA, Legacy Modernization, Public Trust Clearance
- **UX & Research:** UX Discovery, User Testing, Heuristic Evaluation, Accessibility Audits
- **Leadership & Communication:** Stakeholder Management, Client Presentations, Mentoring, Public Speaking
- **Frontend Architecture:** Component Architecture, Design Systems, Responsive Design
- **Delivery:** SDLC, Agile/Scrum, CI/CD Pipelines

TECHNICAL SKILLS

- **AI Tools:** Claude Code, Claude API, Cursor (Agent & Ask modes), Grok, MCP Servers, SKILL.md, CLAUDE.md, AGENTS.md, Midjourney
- **Languages & Frameworks:** TypeScript, JavaScript (ES6+), React, Angular, HTML, CSS, SASS
- **Back End & Data:** Node.js, Express, SQLite (sqlite-vec), Firebase
- **Platforms:** Shopify (Liquid), WordPress, MailerLite
- **Accessibility Testing:** JAWS, NVDA, WAVE, aXe, Lighthouse
- **Design & Workflow:** Figma, Axure, Sketch, Photoshop, Illustrator, Git, GitHub, Bitbucket, Jira

PROFESSIONAL EXPERIENCE

Senior Engineer, Applied AI & Front-End — Zentropolis (Independent) *Alexandria VA (remote)* 01/2025 – Present

UX/UI design and front-end development for early-stage products. From ground-up builds to completing and launching AI-assisted prototypes for founders and startups.

- **Agent Fleet (Applied AI):** Built a multi-agent staff (bookkeeper, property manager, secretary, researcher), each with only the keys it needs: read-only by default, one agent allowed to spend money, and every task orchestrated through a single coordinator agent. Wrote the SKILL.md rulebooks that keep them from wandering off the job and ensure portability to other systems. Connected Claude to Shopify, MailerLite, Metricool, Google Workspace, and Todoist via MCP servers, with store and marketing data strictly read-only. *Tools: Grok, Claude, MCP servers, Google Workspace, Todoist*
- **FitzaCo Opportunity Evaluator (in progress, Nov 2026):** Building a tool that answers one question: is this product idea worth making? It ferrets through product notes and sources like Google Trends (with Amazon sales and Reddit conversations planned) to identify market gaps and then evaluate them for viability in general and in regard to specific business models. Unlike general "find me a good idea" prompts, this has highly defined metrics and guardrails to keep the system on track and useful long term. I measure it like any production system: every claim is cited, it refuses to guess on thin data, and it's backtested against products already launched or killed. *Tools: TypeScript, Node.js, React, Claude API, SQLite*
- **Meshperanto:** Built a word-level Huffman compression codec for Meshtastic LoRa mesh networks, squeezing natural language text messages down by 1.5–2.5x so they travel farther over the tight 237-byte packet limit. Open-sourced it with a React playground for live testing. *Tools: TypeScript, Node.js, React*
- **QuoDrox:** Designed, built, and launched the store for QuoDrox, a line of colorful wooden tiles for tabletop RPGs. Started as an accessible static landing page and waitlist, then moved to Shopify. Rebuilt the store as a fresh Dawn-based theme with a token-driven design system and custom Liquid sections, developed in Cursor. Wired a weekly sales report that Claude builds from read-only Shopify and MailerLite connectors, plus Grok Bot inventory alerts across this site and an Etsy store selling the same products. *Tools: Shopify (Liquid), Cursor, Claude, MCP, MailerLite, Git, Grok*
- **Squirl!:** Took a founder's AI-built prototype of a mobile social app to a launch-ready product: UX consultation to define goals and flows, then refactoring the architecture the AI tools couldn't get right. *Tools: Figma, ReactJS, SASS, Replit, Photoshop, Illustrator*

- B**ackstoryr:** First agentic AI build: a backstory generator for tabletop RPGs. Random tables alone produced incoherent backstories and a raw LLM produced nonsense, so user-selected traits now constrain the model to produce coherent narratives without users having to become prompt engineers. Users can tweak the options and keep regenerating results until they get something they like. *Tools: ReactJS, xAI API, WordPress*
- **urAtlas Odyssey:** First full-stack build owning the whole system: a 12 Week Year-style accountability platform for life coaches, with goal setting, momentum scoring, and check-in dashboards. Architected the React front end myself and directed Cursor through back-end scaffolding, Express routing, and SQLite setup, shipping a complete app in weeks instead of months. *Tools: Figma, ReactJS, Cursor, Express, SQLite, Firebase*
- **QuestRex Character Generato**r: Designed the UX and built the front-end for a complex tabletop RPG character generator managing 50+ interconnected fields with cascading dependencies. Hand-coded the React app, then used Cursor to package it as a WordPress plugin. *Tools: ReactJS, SASS, WordPress, Cursor*

Senior Frontend Engineer — Publicis Sapient *Washington, DC Metro Area* 03/2013 – 12/2024

UX, UI, and front-end developer for a wide variety of government and private sector projects.

- **Consumer Product Safety Commission (CPSC):** Worked with a team to hammer poorly defined internal processes into a streamlined React application. Led UX discovery, produced wireframes and prototypes, conducted accessibility and usability testing (JAWS, NVDA, WAVE) across 40+ custom components, and presented design decisions directly to agency stakeholders. Building accessibility in from the start made the Section 508 compliance audit relatively painless. *Tools: Axure, ReactJS, SASS, JAWS/NVDA/WAVE, Section 508*
- **HRSA Health Workforce Connector:** Defined UX patterns and built a reusable component library deployed across 5+ federal healthcare websites. Guided design through multiple Angular framework rewrites (starting on Angular Beta) while maintaining full WCAG/Section 508 compliance. Mentored team members on accessible component patterns and worked with the UX team to set style and component conventions across the sub-apps. *Tools: Axure, Angular, SASS, WAVE, Section 508*
- **PenFed Credit Union:** Led visual and interaction design consistency for a hybrid mobile banking app across iOS, Android, and legacy devices, ensuring a cohesive usability experience across a fragmented device landscape. *Tools: PhoneGap (Cordova), HTML, CSS, JavaScript*

Lead UX & Senior Front-End Developer — DST / SDSE Washington, DC Metro Area 04/2008 – 01/2011

Led UX and front-end development for federal client engagements.

- **Dept. of Energy HR Website:** Led the redesign of a federal HR portal, wrangling a roomful of stakeholders and bureaucratic requirements into consensus and producing a clean, accessible static front-end. Conducted heuristic evaluation of the legacy site and handled design sign-offs. *Tools: Wireframing, HTML, CSS, JavaScript, Accessibility*

EDUCATION & ADDITIONAL INFORMATION

- **Education**: BA, Communications from Old Dominion University
- **Affiliations:** Toastmasters, The Explorers Club, Philosophical Society of Washington, Freemasons
- **References:** Available upon request